Configure MySQL Databases for WSO2 Identity Server

Introduction

By default, WSO2 Identity Server stores its data in embedded H2 databases. H2 is fine for a laptop, but it is not supported for production. In this lab you move WSO2 Identity Server 7.3.0 to MySQL and keep each kind of data in its own database:

Database	Configured in deployment.toml	What it stores	Script to run
WSO2_IDENTITY_DB	[database.identity_db]	Applications, connections, OAuth2 tokens, sessions, login flows	dbscripts/identity/mysql.sql
WSO2_SHARED_DB	[database.shared_db]	Registry: server and tenant configuration	dbscripts/mysql.sql
WSO2_USER_DB	[database.user] and [realm_manager]	Primary user store: users, groups, roles and their attributes	dbscripts/mysql.sql
WSO2_CONSENT_DB	[datasource.WSO2ConsentDB] and [authentication.consent]	Consent purposes and the consents users give to applications	dbscripts/consent/mysql.sql
WSO2_AGENT_IDENTITY_DB	[datasource.AgentIdentity]	AI agent identities (Lab 04)	dbscripts/identity/agent/mysql.sql

Only WSO2_CARBON_DB stays on H2. It holds only the local registry of each node, so it is normally left on H2.

CC BY 4.0

Important: do this lab on a **new copy** of the WSO2 Identity Server pack (extract the zip into a new folder). The data you created in other labs is in H2 and is not moved to MySQL. Stop any other WSO2 Identity Server that uses port 9443 before you start this one.

Prerequisites

1. MySQL Server 8.0 or later is installed and running on port 3306, and you can log in as `root` with the MySQL client (`mysql`). On Windows, install it with the MySQL Installer and use **MySQL 8.0 Command Line Client** or `mysql.exe`.
2. A new copy of WSO2 Identity Server 7.3.0 is extracted. Its folder is called `<IS_HOME>` below.
3. JDK 21 or 25 is installed and `JAVA_HOME` is set.
4. Apache Tomcat with `pickup-dispatch.war` deployed (Lab 05), for testing the consent database.

Step 1: Create the databases and a database user

1. Open the MySQL client as root:

   ```
   mysql -u root -p
   ```

2. Run the following statements. They create the five databases and a user `wso2user` that WSO2 Identity Server uses to connect:

   ```
   CREATE DATABASE WSO2_IDENTITY_DB CHARACTER SET latin1;
   CREATE DATABASE WSO2_SHARED_DB CHARACTER SET latin1;
   CREATE DATABASE WSO2_USER_DB CHARACTER SET latin1;
   CREATE DATABASE WSO2_CONSENT_DB CHARACTER SET latin1;
   CREATE DATABASE WSO2_AGENT_IDENTITY_DB CHARACTER SET latin1;

   CREATE USER 'wso2user'@'localhost' IDENTIFIED BY 'Wso2@123';
   GRANT ALL PRIVILEGES ON WSO2_IDENTITY_DB.* TO 'wso2user'@'localhost';
   GRANT ALL PRIVILEGES ON WSO2_SHARED_DB.* TO 'wso2user'@'localhost';
   ```

CC BY 4.0

```
GRANT ALL PRIVILEGES ON WSO2_USER_DB.* TO 'wso2user'@'localhost';
GRANT ALL PRIVILEGES ON WSO2_CONSENT_DB.* TO 'wso2user'@'localhost';
GRANT ALL PRIVILEGES ON WSO2_AGENT_IDENTITY_DB.* TO
'wso2user'@'localhost';
FLUSH PRIVILEGES;
```

Why CHARACTER SET latin1? MySQL 8 creates databases with `utf8mb4` by default. Most tables in the WSO2 identity script set `latin1` themselves, but a few do not, so they would get `utf8mb4`. MySQL then refuses a foreign key between the two and the script stops with `ERROR 3780 ... IDN_OAUTH2_REFRESH_TOKEN_BINDING_ibfk_1 are incompatible`. Creating the databases with `latin1` keeps all tables consistent. User attribute values (such as names) are still stored as `utf8mb4`, because the user store script sets that column explicitly.

Step 2: Create the tables

1. In the same MySQL client, run the WSO2 scripts against each database. Replace `<IS_HOME>` with the full path of your pack and use forward slashes in the path, also on Windows (for example `C:/wso2/wso2is-7.3.0`):

```
USE WSO2_IDENTITY_DB;
SOURCE <IS_HOME>/dbscripts/identity/mysql.sql;

USE WSO2_SHARED_DB;
SOURCE <IS_HOME>/dbscripts/mysql.sql;

USE WSO2_USER_DB;
SOURCE <IS_HOME>/dbscripts/mysql.sql;

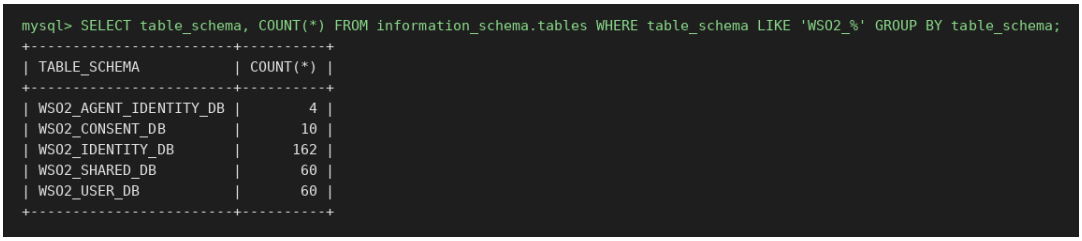
USE WSO2_CONSENT_DB;
SOURCE <IS_HOME>/dbscripts/consent/mysql.sql;

USE WSO2_AGENT_IDENTITY_DB;
SOURCE <IS_HOME>/dbscripts/identity/agent/mysql.sql;
```

CC BY 4.0

2. The shared and user databases use the same script: it contains both the registry tables and the user store tables. WSO2 Identity Server uses the registry tables in WSO2_SHARED_DB and the user store tables in WSO2_USER_DB.
3. Check that all five databases have tables. You should see 4, 10, 162, 60 and 60 tables:

```
SELECT table_schema, COUNT(*) FROM information_schema.tables
WHERE table_schema LIKE 'WSO2_%' GROUP BY table_schema;
```



CC BY 4.0

4. Type `exit` to leave the MySQL client.

Step 3: Add the MySQL JDBC driver

1. Download MySQL Connector/J from https://dev.mysql.com/downloads/connector/j/ (choose **Platform Independent**) and extract it.
2. Copy `mysql-connector-j-<version>.jar` into `<IS_HOME>/repository/components/lib`.

Step 4: Point WSO2 Identity Server to the databases

1. Open `<IS_HOME>/repository/conf/deployment.toml` in a text editor.
2. **Replace** the existing `[database.identity_db]`, `[database.shared_db]` and `[datasource.AgentIdentity]` sections (they point to H2) with the following. Keep the other sections as they are.

```
[database.identity_db]
type = "mysql"
url = "jdbc:mysql://localhost:3306/WSO2_IDENTITY_DB?
useSSL=false&amp;allowPublicKeyRetrieval=true"
username = "wso2user"
password = "Wso2@123"
driver = "com.mysql.cj.jdbc.Driver"

[database.shared_db]
type = "mysql"
url = "jdbc:mysql://localhost:3306/WSO2_SHARED_DB?
useSSL=false&amp;allowPublicKeyRetrieval=true"
username = "wso2user"
password = "Wso2@123"
driver = "com.mysql.cj.jdbc.Driver"

[database.user]
type = "mysql"
```

CC BY 4.0

```
url = "jdbc:mysql://localhost:3306/WSO2_USER_DB?
useSSL=false&amp;allowPublicKeyRetrieval=true"
username = "wso2user"
password = "Wso2@123"
driver = "com.mysql.cj.jdbc.Driver"

[realm_manager]
data_source = "WSO2USER_DB"

[datasource.WSO2ConsentDB]
id = "WSO2CONSENT_DB"
url = "jdbc:mysql://localhost:3306/WSO2_CONSENT_DB?
useSSL=false&amp;allowPublicKeyRetrieval=true"
username = "wso2user"
password = "Wso2@123"
driver = "com.mysql.cj.jdbc.Driver"
jmx_enable = false

[authentication.consent]
data_source = "jdbc/WSO2CONSENT_DB"

[datasource.AgentIdentity]
id = "AgentIdentity"
url = "jdbc:mysql://localhost:3306/WSO2_AGENT_IDENTITY_DB?
useSSL=false&amp;allowPublicKeyRetrieval=true"
username = "wso2user"
password = "Wso2@123"
driver = "com.mysql.cj.jdbc.Driver"
```

3. Under the existing `[user_store]` section (which has `type = "database_unique_id"`), add the following. MySQL already compares usernames without case, so WSO2 Identity Server does not need to do it again (this is the recommendation in the WSO2 documentation):

```
[user_store.properties]
CaseInsensitiveUsername = false
UseCaseSensitiveUsernameForCacheKeys = false
```

4. Save the file.

CC BY 4.0

What each part does:

1. `[database.identity_db]` and `[database.shared_db]` move the identity data and the registry to MySQL.
2. `[database.user]` creates a datasource called `WSO2USER_DB`. `[realm_manager] data_source` makes the primary user store use it instead of the shared database.
3. `[datasource.WSO2ConsentDB]` creates a datasource with the JNDI name `jdbc/WSO2CONSENT_DB`, and `[authentication.consent] data_source` tells the consent manager to use it. Without this, consents are stored in the identity database.
4. Write `&` in the URL as `&`. The URL is copied into an XML file when the server starts, and a plain `&` makes the server fail with `Unexpected character '=' ... expected a semi-colon after the reference for entity 'allowPublicKeyRetrieval'`.
5. `useSSL=false` and `allowPublicKeyRetrieval=true` are for this lab only: MySQL runs on the same machine. In production, enable SSL to the database server and store the password with the cipher tool (`$secret{...}`).
6. `[datasource.AgentIdentity]` keeps its id `AgentIdentity` (WSO2 Identity Server looks up `jdbc/AgentIdentity` for agents) and now points to WSO2_AGENT_IDENTITY_DB.
7. The consent datasource uses the same `[datasource.<name>]` form as AgentIdentity. Do not write it as `[[datasource]]`: TOML does not allow both forms in the same file.

Step 5: Start the server

1. Start WSO2 Identity Server from `<IS_HOME>/bin`: run `wso2server.bat` on Windows or `sh wso2server.sh` on Linux and macOS.

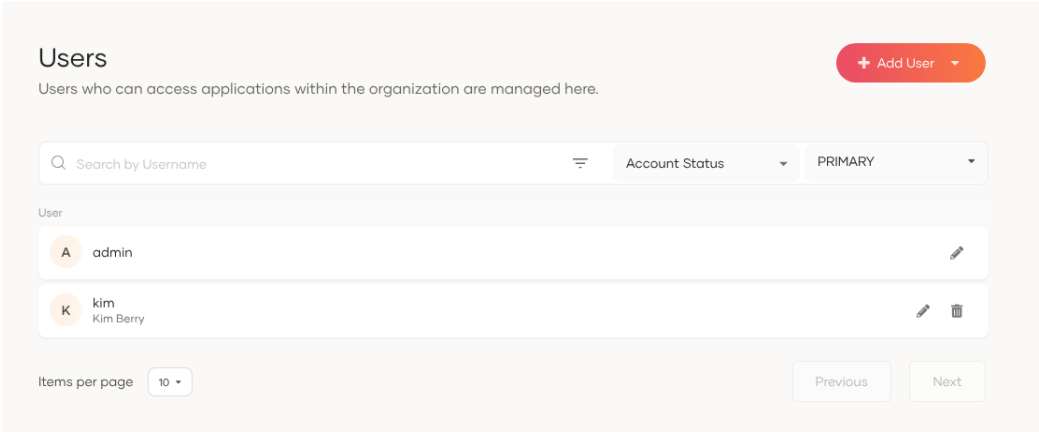
CC BY 4.0

2. Wait for `WSO2 Carbon started`. Check that no `ERROR` lines are printed. On the first start, WSO2 Identity Server creates the `admin` user and the default configuration in the MySQL databases.

Try It: check that each database is used

A. User database

1. Log in to the Console (`https://localhost:9443/console`) as `admin` / `admin`. This works only if the admin user was created in WSO2_USER_DB.
2. Go to **User Management > Users**, click **Add User > Single User** and add the user `kim` (Email `kim@example.com`, First Name `Kim`, Last Name `Berry`). Select **Set a password for the user**, enter `Kim@12345` and finish the wizard.



CC BY 4.0

3. In the MySQL client, check that both users are in WSO2_USER_DB:

```
SELECT UM_USER_NAME FROM WSO2_USER_DB.UM_USER;
```

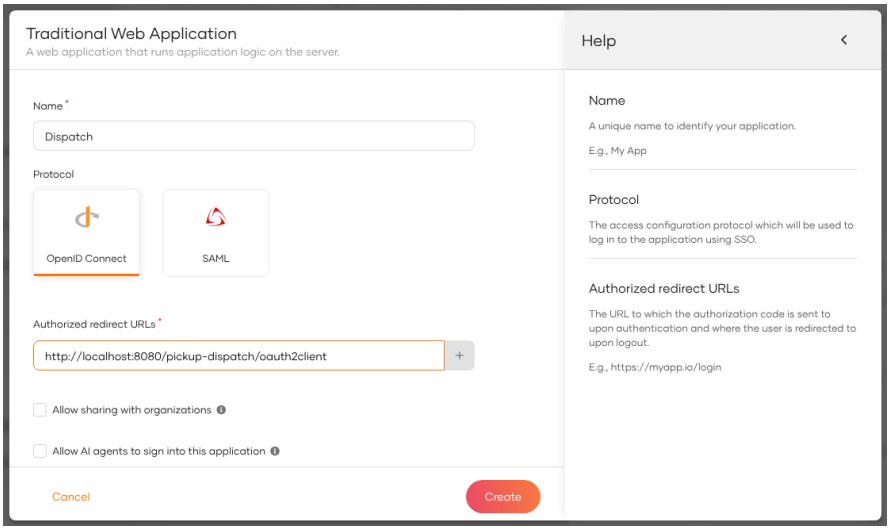
```
mysql> SELECT UM_USER_NAME FROM WSO2_USER_DB.UM_USER;
+--------------+
| UM_USER_NAME |
+--------------+
| admin        |
| kim          |
+--------------+
```

CC BY 4.0

Groups are stored in the `UM_ROLE` table and roles in the `UM_HYBRID_ROLE` table of the same database.

B. Identity database: register an application

1. Go to **Applications**, click **+ New Application** and select **Traditional Web Application**.
2. Enter the name `Dispatch`, select **OpenID Connect**, enter `http://localhost:8080/pickup-dispatch/oauth2client` as the **Authorized redirect URL** and click **Create**.



CC BY 4.0

3. On the **Protocol** tab, copy the **Client ID** and **Client secret**. Put them in `consumerKey` and `consumerSecret` in `<TOMCAT_HOME>/webapps/pickup-dispatch/WEB-INF/classes/dispatch.properties` and restart Tomcat (as in Lab 05).
4. Check that the application is in WSO2_IDENTITY_DB:

```
SELECT APP_NAME FROM WSO2_IDENTITY_DB.SP_APP;
```

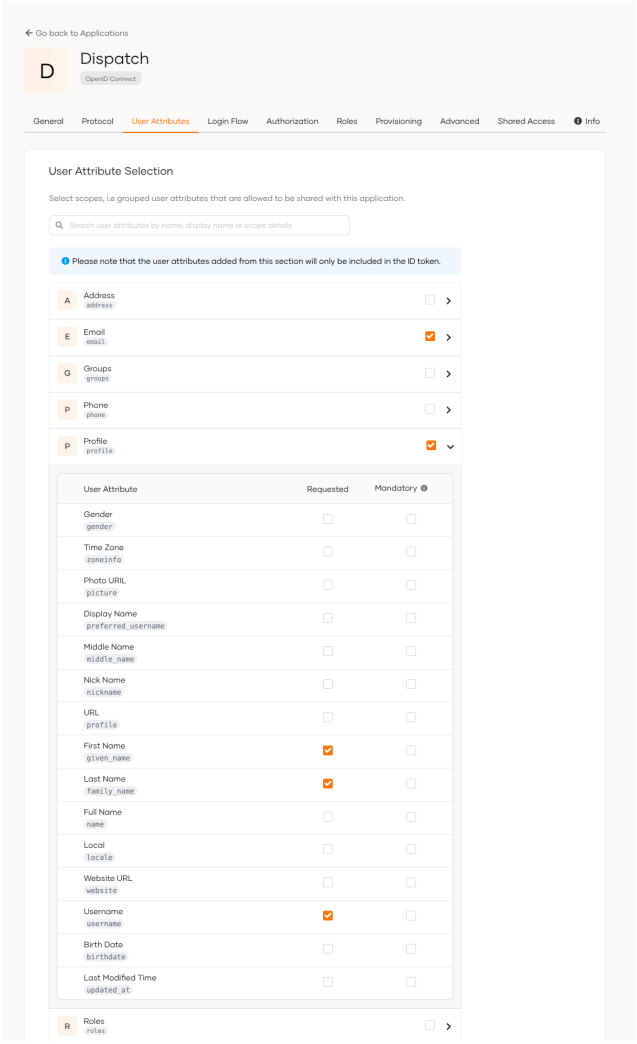
```
mysql> SELECT APP_NAME FROM WSO2_IDENTITY_DB.SP_APP;
+---------------------+
| APP_NAME            |
+---------------------+
| Console             |
| Dispatch            |
| My Account          |
| wso2carbon-local-sp |
+---------------------+
```

CC BY 4.0

C. Consent database: configure consent on the application

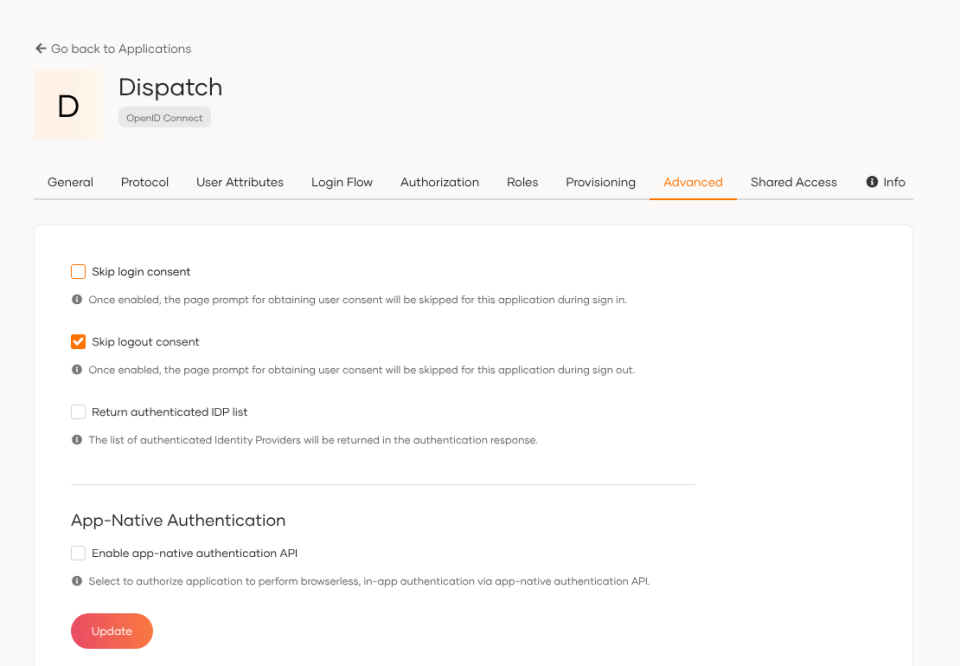
A consent is saved only when the application asks for user attributes and the user is shown the consent page. Configure both on the Dispatch application:

1. Go to the **User Attributes** tab. Tick the **Email** scope. Expand **Profile** and tick **Requested** for **First Name** and **Last Name**. Click **Update**.



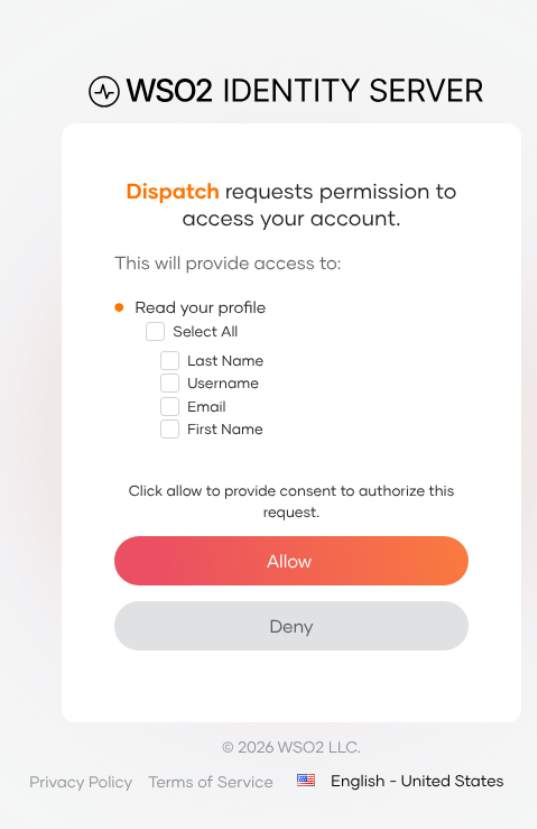
CC BY 4.0

2. Go to the **Advanced** tab. Untick **Skip login consent** and click **Update**. This option is ticked by default for new applications; while it is ticked, users are never asked for consent and nothing is saved in WSO2_CONSENT_DB.



CC BY 4.0

3. Open `http://localhost:8080/pickup-dispatch` in a private window, click **Login** and sign in as `kim`.
4. WSO2 Identity Server shows the consent page with the attributes that Dispatch requests. Click **Allow**.



CC BY 4.0

5. Check that Kim's consent for Dispatch is saved in WSO2_CONSENT_DB:

```
SELECT r.PII_PRINCIPAL_ID, s.SP_NAME, r.STATE
FROM WSO2_CONSENT_DB.CM_RECEIPT r
JOIN WSO2_CONSENT_DB.CM_RECEIPT_SP_ASSOC s ON s.CONSENT_RECEIPT_ID =
r.CONSENT_RECEIPT_ID;
```



CC BY 4.0

D. Agent identity database

1. In the Console, go to **Agents**, click **+ New Agent**, enter `Travel orchestrator` as the **Agent Name** and click **Create**. Click **Done**.
2. Check that the agent is stored in WSO2_AGENT_IDENTITY_DB (the agent ID is a generated value):

```
SELECT UM_USER_NAME AS AGENT_ID FROM WSO2_AGENT_IDENTITY_DB.UM_USER;
```

```
mysql> SELECT UM_USER_NAME AS AGENT_ID FROM WSO2_AGENT_IDENTITY_DB.UM_USER;
+--------------------------------------+
| AGENT_ID                             |
+--------------------------------------+
| f5754490-ed9a-4544-a44a-2d41aadb5747 |
+--------------------------------------+
```

CC BY 4.0

E. Shared database

1. Check that the registry data is in WSO2_SHARED_DB (the number of rows can be different):

```
SELECT COUNT(*) AS REG_RESOURCE_ROWS FROM WSO2_SHARED_DB.REG_RESOURCE;
```

```
mysql> SELECT COUNT(*) AS REG_RESOURCE_ROWS FROM WSO2_SHARED_DB.REG_RESOURCE;
+-------------------+
| REG_RESOURCE_ROWS |
+-------------------+
|               291 |
+-------------------+
```

CC BY 4.0

F. The data survives a restart

1. Stop WSO2 Identity Server (Ctrl+C) and start it again.
2. Log in to My Account (`https://localhost:9443/myaccount`) as `kim`, and check that the Dispatch application and the Travel orchestrator agent are still listed in the Console. All this data is now in MySQL, not in H2.

CC BY 4.0

Troubleshooting

Error	Cause and fix
ERROR 3780 ... IDN_OAUTH2_REFRESH_TOKEN_BINDING_ibfk_1 are incompatible (while running the identity script)	The database was created with the default utf8mb4 character set. Drop it, create it again with CHARACTER SET latin1 (Step 1) and run the script again.
Unexpected character '=' ... expected a semi-colon after the reference for entity 'allowPublicKeyRetrieval'	A plain & in a database URL in deployment.toml. Write it as &.
Communications link failure / Connection refused	MySQL is not running, or the host or port in the URL is wrong.
Public Key Retrieval is not allowed	allowPublicKeyRetrieval=true is missing from the URL.
Access denied for user 'wso2user'@'localhost'	Wrong username or password in deployment.toml, or the GRANT statements were not run.
ClassNotFoundException: com.mysql.cj.jdbc.Driver	The Connector/J JAR is not in <IS_HOME>/repository/components/lib, or the server was not restarted after copying it.
No consent page at login, and CM_RECEIPT stays empty	Skip login consent is still ticked on the Advanced tab of the application, or the application does not request any user attributes (Try It, part C).

CC BY 4.0